

Rohan Surana

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Summary

Researcher at Qualcomm working on proactive, personalized LLM agents and long-horizon context management and memory. Published research on RL post-training, preference optimization, and agent evaluation at NeurIPS, ICML, COLM, ICLR, and ACL.

Education

Masters of Science in Data Science

University of California, San Diego (UCSD) GPA: 3.88/4.00. Advisor: Julian McAuley

March 2026

San Diego, CA

Bachelor of Science in Software Engineering

San Jose State University, GPA: 3.87/4.00 (Summa cum laude)

May 2022

San Jose, CA

Experience

Qualcomm AI Research

Research Scientist

San Diego, CA

Apr 2026 – Present

- Research on proactive agents and long-range context management to improve agentic capabilities

Dell Technologies

AI Research Intern

Hopkinton, MA

June 2025 – Sept 2025

- Built multi-agent LLM system (LangGraph/LangChain + vLLM) with request batching and KV-cache, cutting p95 latency by 40%
- Designed scalable RAG pipelines integrating LLMs with vector databases using hybrid search and intelligent chunking strategies
- Engineered short/long-term memory with retrieval caching to reduce serving cost & exposed APIs for integration with existing workflows
- Developed agent-based monitoring using MCP and A2A protocols to automate telemetry collection, anomaly detection, and remediation

Dell Technologies

Software Engineer II

Santa Clara, CA

Mar 2024 – Aug 2024

- Architected TOSCA-based framework to orchestrate resources, cutting provisioning time 30% and improving infrastructure flexibility
- Built real-time infrastructure digital twin with predictive analytics using graph databases to enable drift detection and safe remediation

Software Engineer I

Jul 2022 – Mar 2024

- Accelerated cluster time-to-ready by 20% through automated Kubernetes operators supporting GPU workloads, KServe, and TorchServe
- Enhanced service performance by redesigning API and proto services with gRPC, leading a 4-intern team to boost efficiency 15%

Selected Publications (full list on [Google Scholar](#))

- Reasoning Pathologies in Large Language Models: A Diagnostic Perspective**
Rohan Surana, J. Wu, S. Yu, G. Mundada, X. Jiang, Z. Huang, R. A. Rossi, J. McAuley, T. Yu | *NeurIPS 2026*
- AMPS: Adaptive Modality Preference Steering via Functional Entropy**
Z. Huang, X. Li, Rohan Surana, T. Yu, R. Wang, J. McAuley, J. Shang, J. Wu | *NeurIPS 2026*
- WS-GRPO: Weakly-Supervised Group-Relative Policy Optimization for Rollout-Efficient Reasoning**
G. Mundada*, Z. Huang*, Rohan Surana*, S. Yu, J. Y. Zhang, X. Li, T. Yu, L. Yao, J. Shang, J. McAuley, J. Wu | *ICML 2026*
- Evaluating Language Model Pluralism through In-the-wild Crowd Discussions**
G. Mundada, Rohan Surana, N. Swaminathan, B. P. Majumder, J. Wu, J. McAuley, Z. Xie | *ACL 2026*
- Importance Sampling for Multi-Negative Multimodal Direct Preference Optimization**
X. Li, C. Wang, J. Wu, Rohan Surana, T. Yu, J. McAuley, J. Shang | *ICLR 2026*
- In-context Ranking Preference Optimization**
J. Wu*, Rohan Surana*, Z. Xie, Y. Shen, Y. Xia, T. Yu, R. Rossi, P. Ammanabrolu, J. McAuley | *COLM 2025*
- Image Difference Captioning via Adversarial Preference Optimization**
Z. Huang, J. Wu, Rohan Surana, T. Yu, D. Arbour, R. Sinha, J. McAuley | *EMNLP 2025*
- MASS-DPO: Multi-Negative Active Sample Selection for Direct Policy Optimization**
Rohan Surana*, X. Li, S. Yu, Y. J. Shen, C. Wang, T. Yu, P. Ammanabrolu, J. Shang, J. McAuley, J. Wu | *Under Review*
- F-GRPO: Factorized Group-Relative Policy Optimization for Unified Candidate Generation and Ranking**
Rohan Surana, G. Mundada, J. Wu, X. Li, Y. Jiao, B. Jin, S. Zhou, T. Yu, R. Sinha, J. Han, J. Shang, J. McAuley | *Under Review*
- Skill-R1: Agent Skill Evolution via Reinforcement Learning**
Y. Vishe, Rohan Surana, X. Jiang, Z. Huang, X. Li, N. L. Kuang, T. Yu, R. A. Rossi, J. Shang, J. McAuley, J. Wu | *Under Review*
- MusiCRS: Benchmarking Audio-Centric Conversational Recommendation**
Rohan Surana*, A. Namburi*, G. Mundada*, A. Lal*, Z. Novack, J. McAuley, J. Wu | *ICASSP 2026*
- An Empirical Study on Zero-Data Bootstrapping for Conversational Recommender Systems**
Rohan Surana*, J. Wu*, Z. Xie*, Y. Xia, N. Kallus, J. McAuley | *EMNLP 2026 (Findings)*

Technical Skills

ML & Agents: PyTorch, Transformers, TensorFlow, scikit-learn, Hugging Face, vLLM, LangGraph

Engineering: Python, Go, Java, SQL, Kubernetes, Docker, FastAPI, gRPC, AWS, GCP